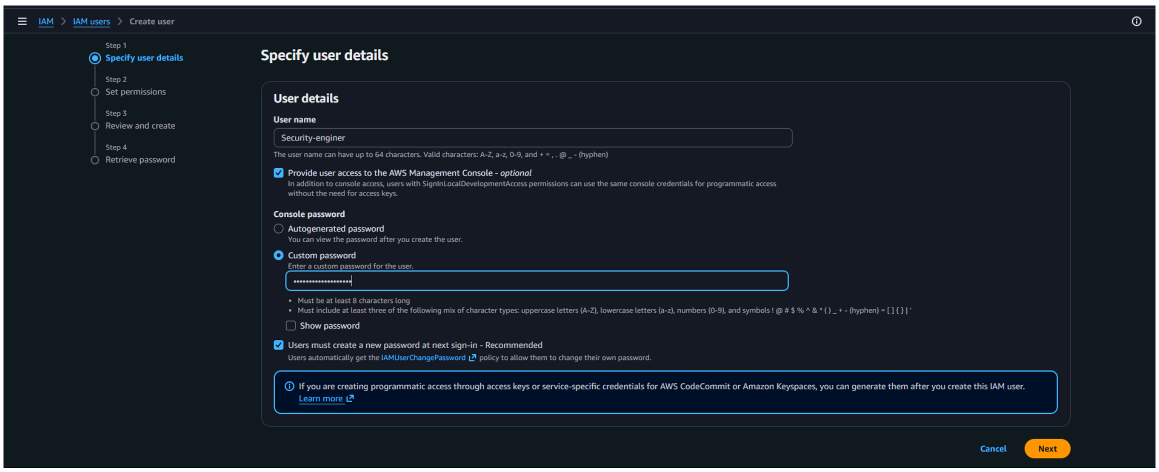
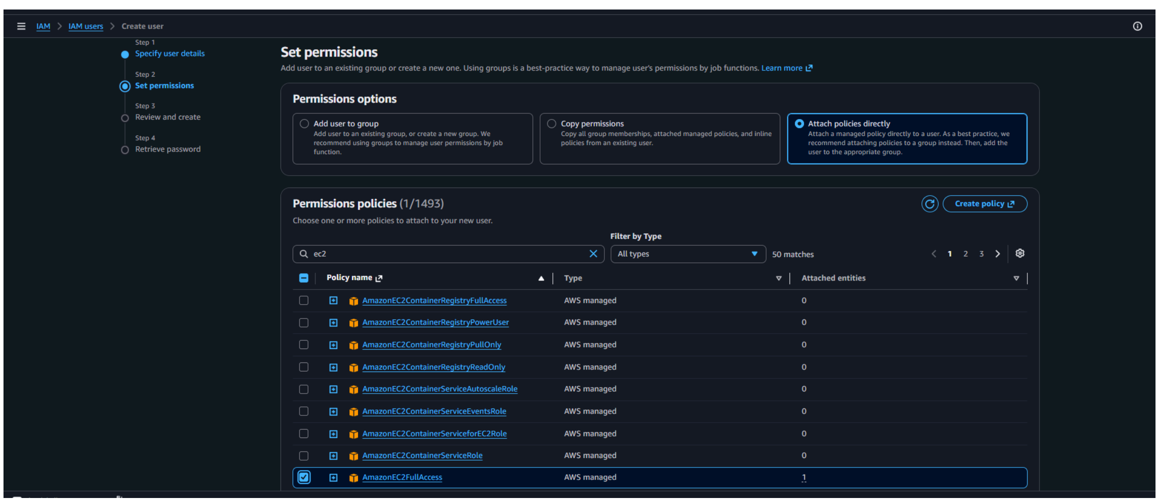
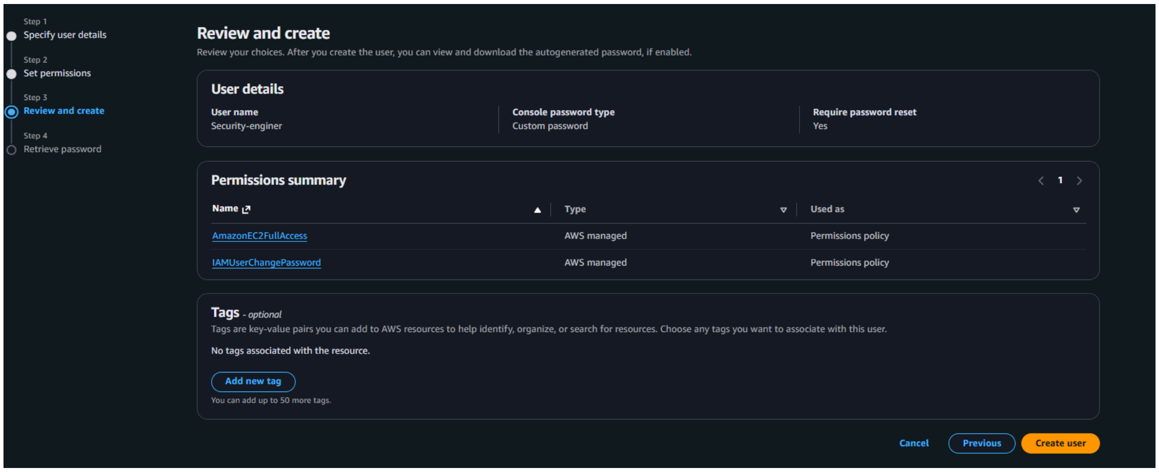
AWS SOC Monitoring - CreateUser Detection Workflow

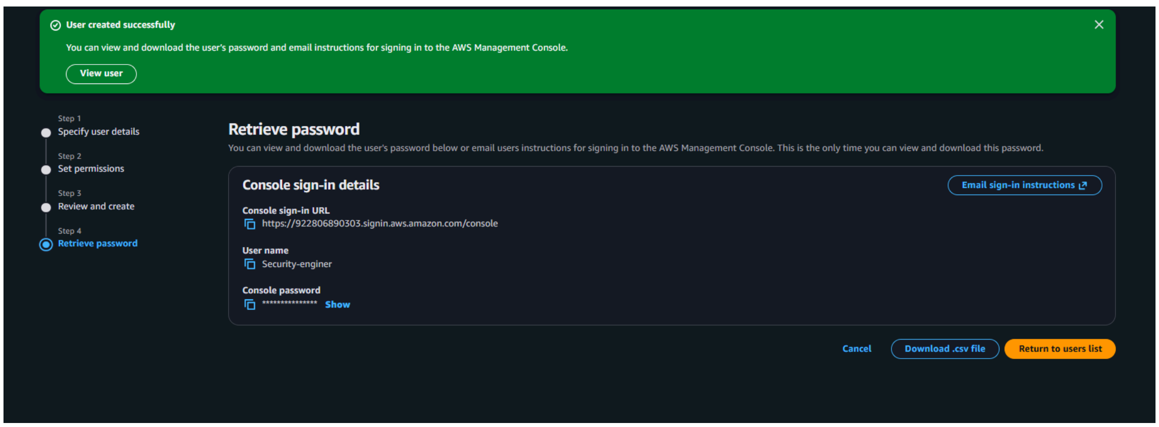
Step 1: Simulate Suspicious Activity

Create a new IAM user inside AWS IAM service. This generates a CreateUser API event in AWS CloudTrail logs.



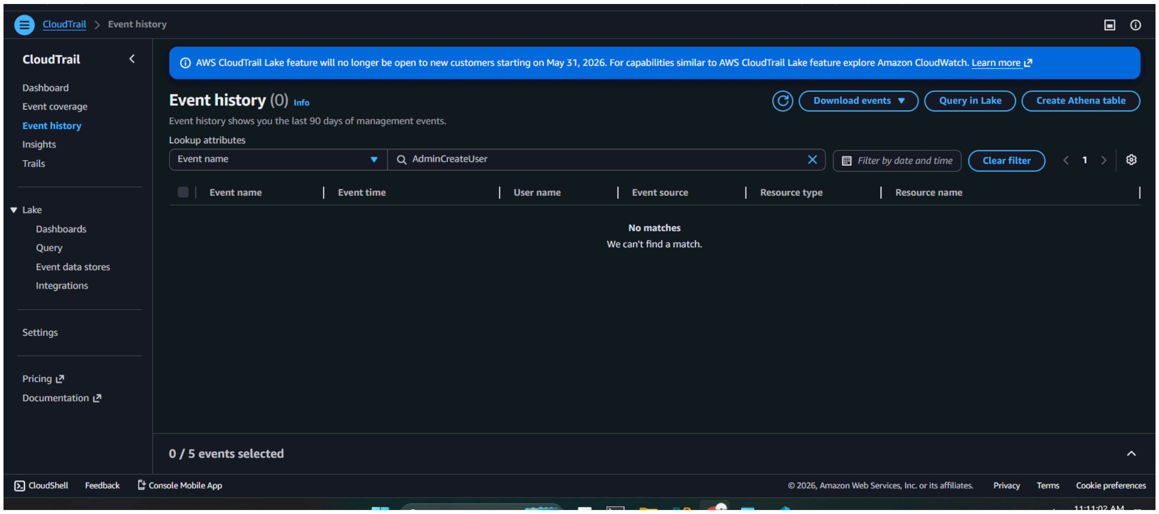


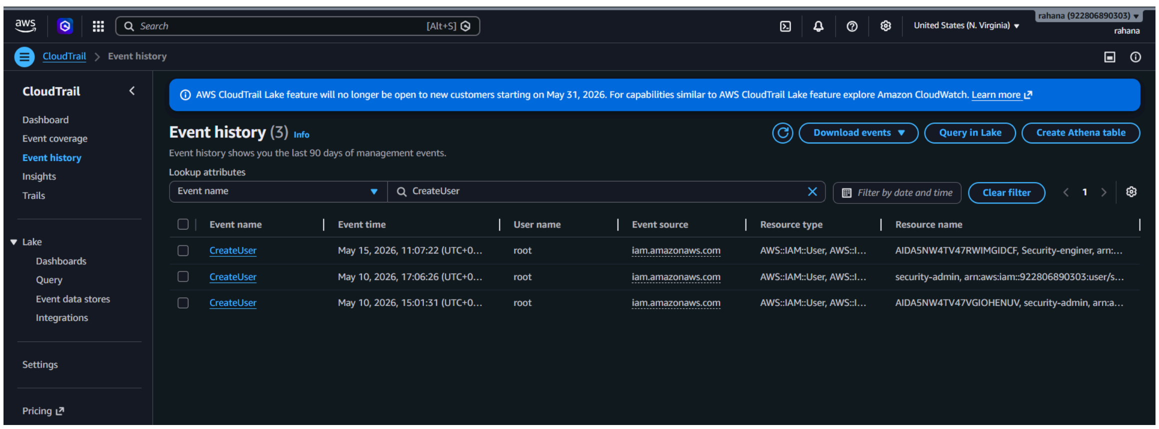

Step 1
Specify user details
Step 2
Set permissions
Step 3
Review and create
Step 4
Retrieve password
Review and create
Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.
User details
User name
Security-enginer
Console password type
Custom password
Require password reset
Yes
Permissions summary
1
Name
Type
Used as
AmazonEC2FullAccess
AWS managed
Permissions policy
IAMUserChangePassword
AWS managed
Permissions policy
Tags - optional
Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.
No tags associated with the resource.
Add new tag
You can add up to 50 more tags.
Cancel
Previous
Create user


User created successfully
You can view and download the user's password and email instructions for signing in to the AWS Management Console.
View user
Step 1
Specify user details
Step 2
Set permissions
Step 3
Review and create
Step 4
Retrieve password
Retrieve password
You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.
Console sign-in details
Email sign-in instructions
Console sign-in URL
https://922806890303.signin.aws.amazon.com/console
User name
Security-enginer
Console password
****
Show
Cancel
Download .csv file
Return to users list

Step 2: Verify CloudTrail Event

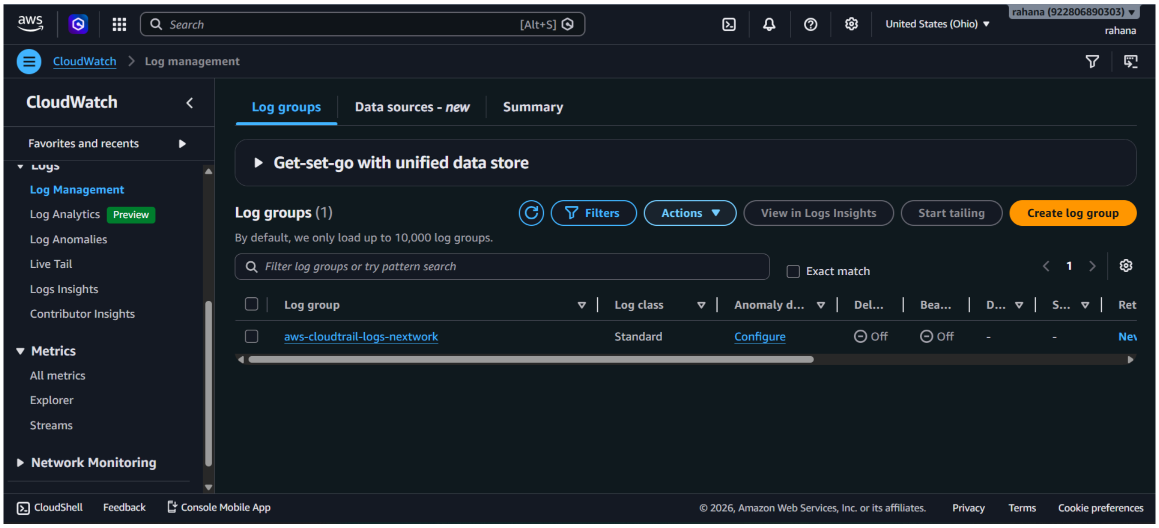
Verify that the CreateUser event is successfully recorded inside AWS CloudTrail Event History.

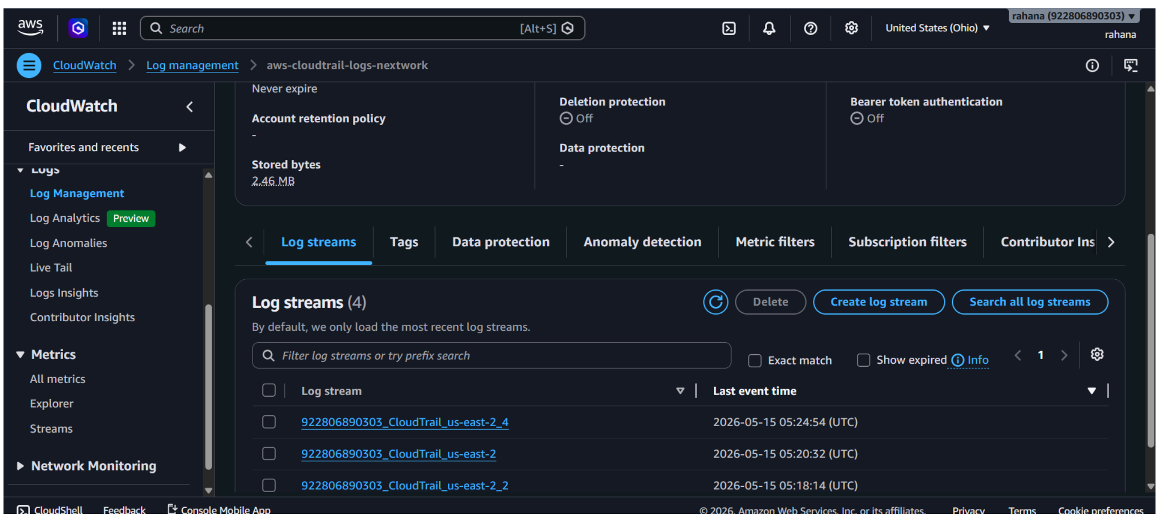




Step 3: Open CloudWatch Log Groups

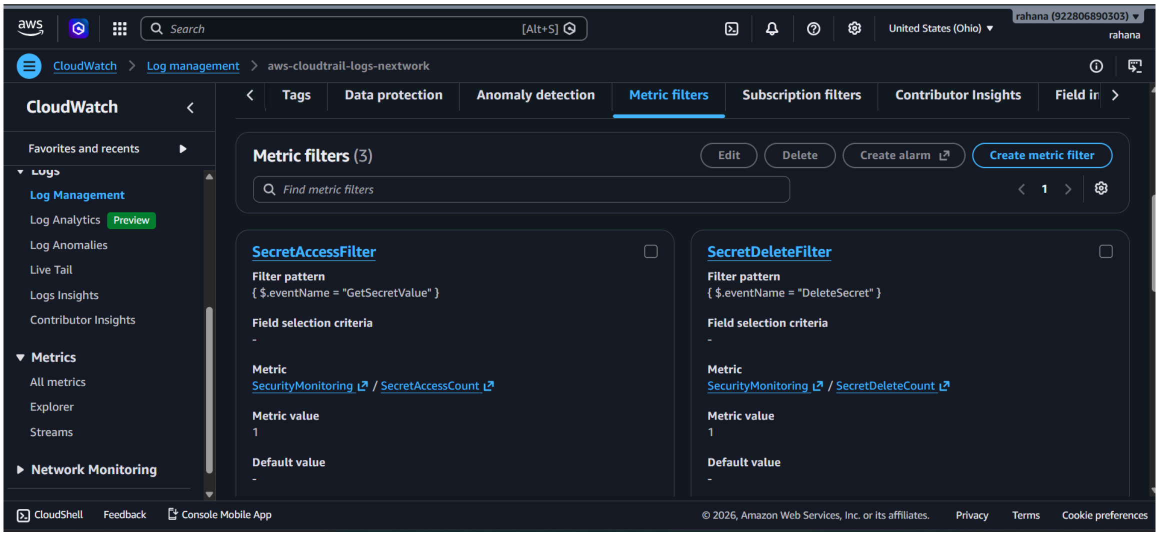
Access CloudWatch Log Groups connected to CloudTrail and verify that CreateUser logs are available in log streams.

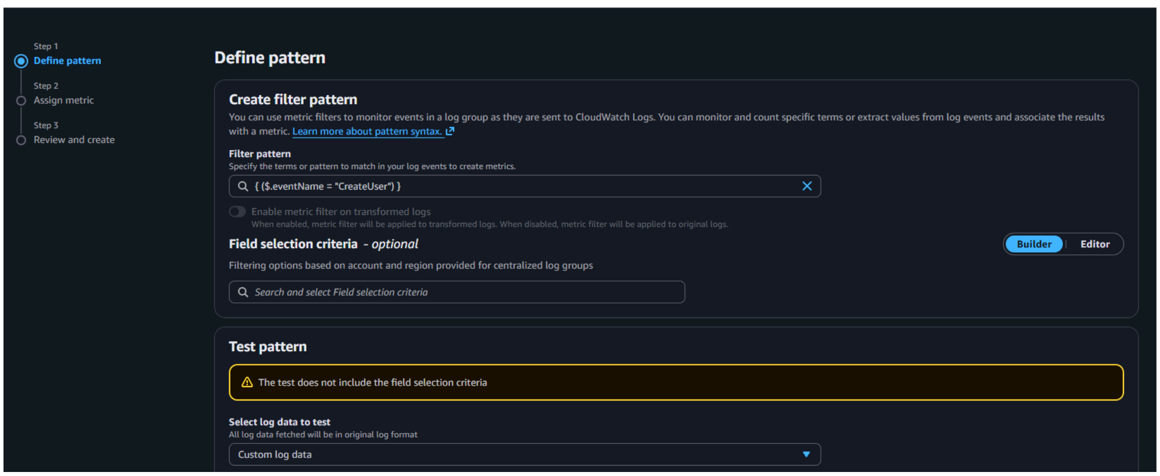


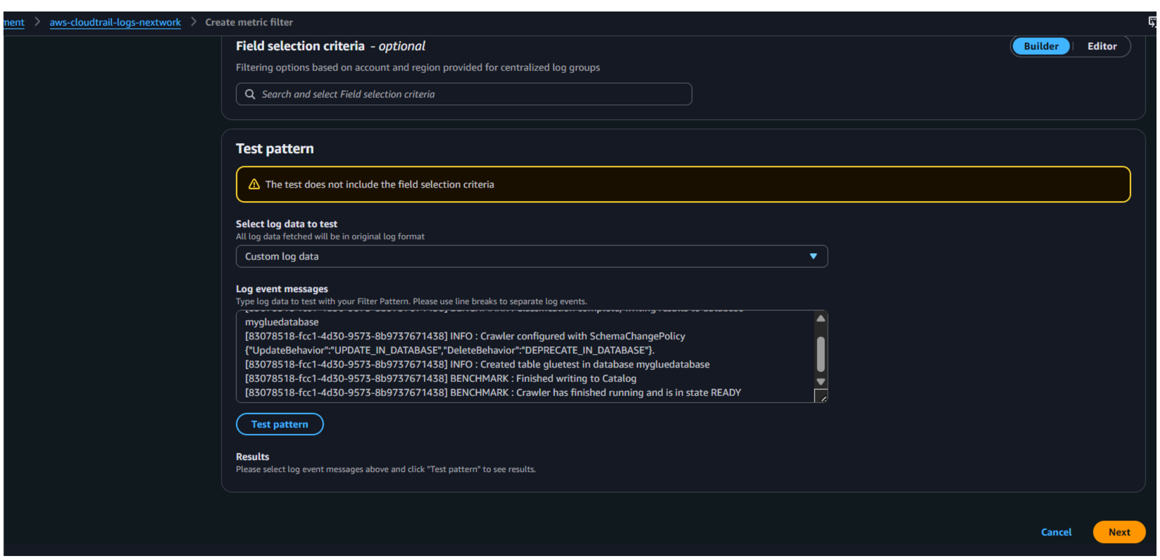


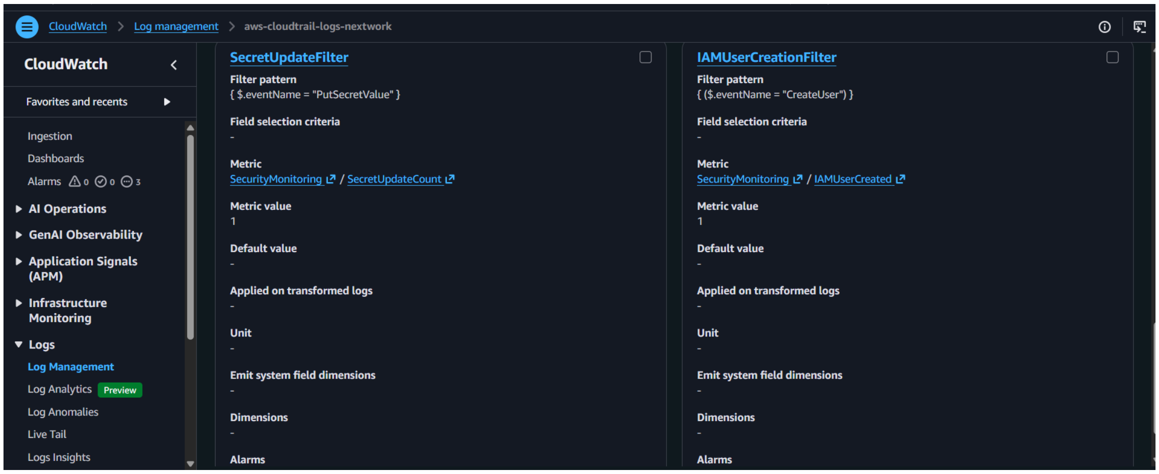
Step 4: Create Metric Filter

Create a CloudWatch Metric Filter using the pattern: { ($.eventName = "CreateUser") } to detect IAM user creation events.


aws
Search
[Alt+S]
United States (Ohio)
rahana (922806890303)
rahana
CloudWatch
Log management
aws-cloudtrail-logs-nextwork
CloudWatch
Tags
Data protection
Anomaly detection
Metric filters
Subscription filters
Contributor Insights
Field in
Favorites and recents
Logs
Log Management
Log Analytics
Preview
Log Anomalies
Live Tail
Logs Insights
Contributor Insights
Metrics
All metrics
Explorer
Streams
Network Monitoring
Metric filters (3)
Edit
Delete
Create alarm
Create metric filter
Find metric filters
1
SecretAccessFilter
Filter pattern
{ $.eventName = "GetSecretValue" }
Field selection criteria
-
Metric
SecurityMonitoring / SecretAccessCount
Metric value
1
Default value
-
SecretDeleteFilter
Filter pattern
{ $.eventName = "DeleteSecret" }
Field selection criteria
-
Metric
SecurityMonitoring / SecretDeleteCount
Metric value
1
Default value
-
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

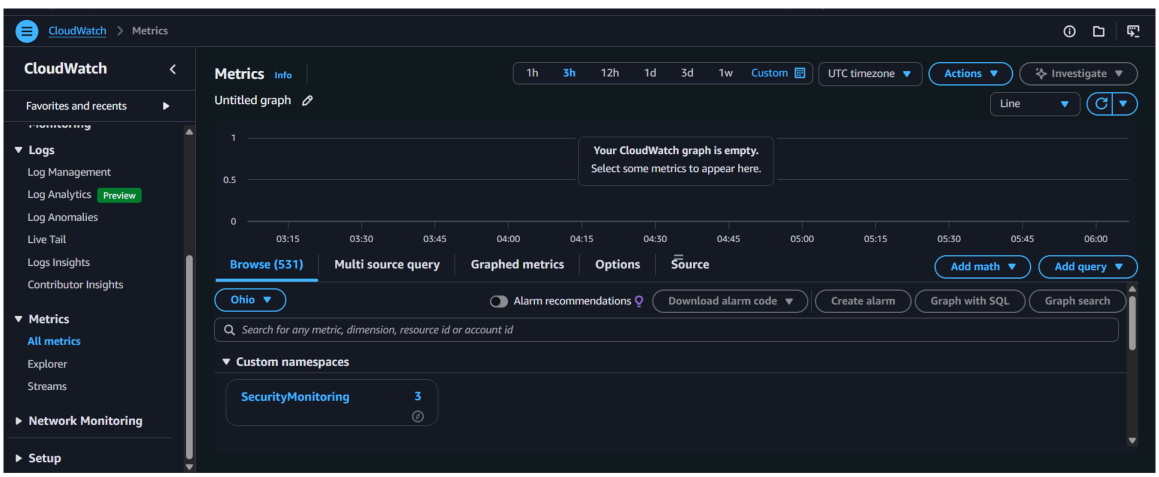

Step 1
Define pattern
Step 2
Assign metric
Step 3
Review and create
Define pattern
Create filter pattern
You can use metric filters to monitor events in a log group as they are sent to CloudWatch Logs. You can monitor and count specific terms or extract values from log events and associate the results with a metric. Learn more about pattern syntax.
Filter pattern
Specify the terms or pattern to match in your log events to create metrics.
{ ($.eventName = "CreateUser") }
Enable metric filter on transformed logs
When enabled, metric filter will be applied to transformed logs. When disabled, metric filter will be applied to original logs.
Field selection criteria - optional
Builder
Editor
Filtering options based on account and region provided for centralized log groups
Search and select Field selection criteria
Test pattern
The test does not include the field selection criteria
Select log data to test
All log data fetched will be in original log format
Custom log data

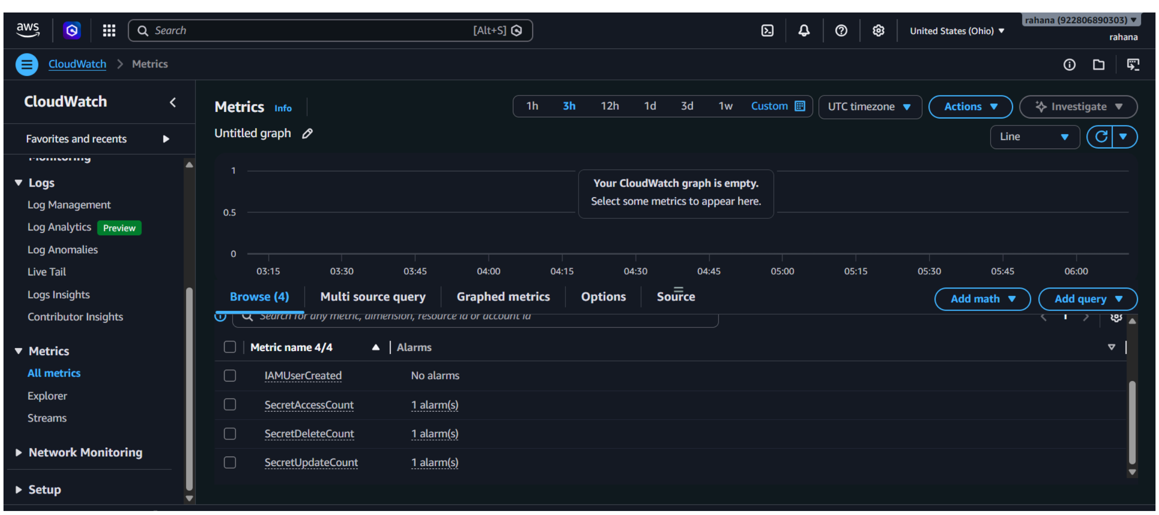

aws-cloudtrail-logs-nextwork
Create metric filter
Field selection criteria - optional
Builder
Editor
Filtering options based on account and region provided for centralized log groups
Search and select Field selection criteria
Test pattern
The test does not include the field selection criteria
Select log data to test
All log data fetched will be in original log format
Custom log data
Log event messages
Type log data to test with your Filter Pattern. Please use line breaks to separate log events.
mygluedatabase
[83078518-fcc1-4d30-9573-8b9737671438] INFO : Crawler configured with SchemaChangePolicy
{"UpdateBehavior":"UPDATE_IN_DATABASE","DeleteBehavior":"DEPRECATE_IN_DATABASE"}.
[83078518-fcc1-4d30-9573-8b9737671438] INFO : Created table gluetest in database mygluedatabase
[83078518-fcc1-4d30-9573-8b9737671438] BENCHMARK : Finished writing to Catalog
[83078518-fcc1-4d30-9573-8b9737671438] BENCHMARK : Crawler has finished running and is in state READY
Test pattern
Results
Please select log event messages above and click "Test pattern" to see results.
Cancel
Next



Step 5: Verify Metric Creation

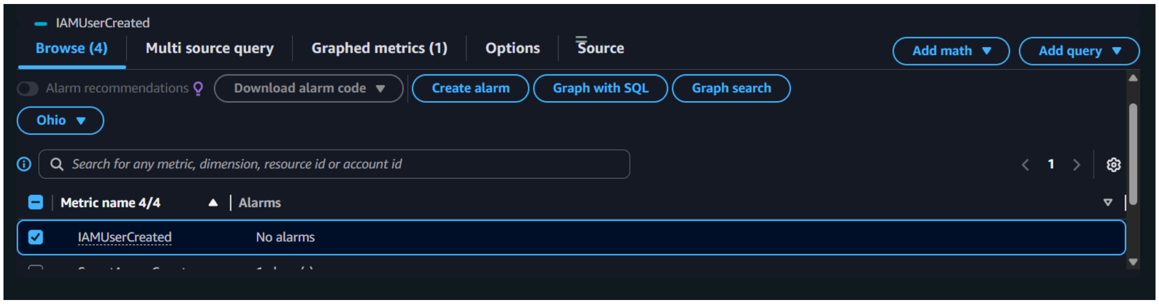
Verify that the IAMUserCreated metric is successfully created inside the SecurityMonitoring namespace.

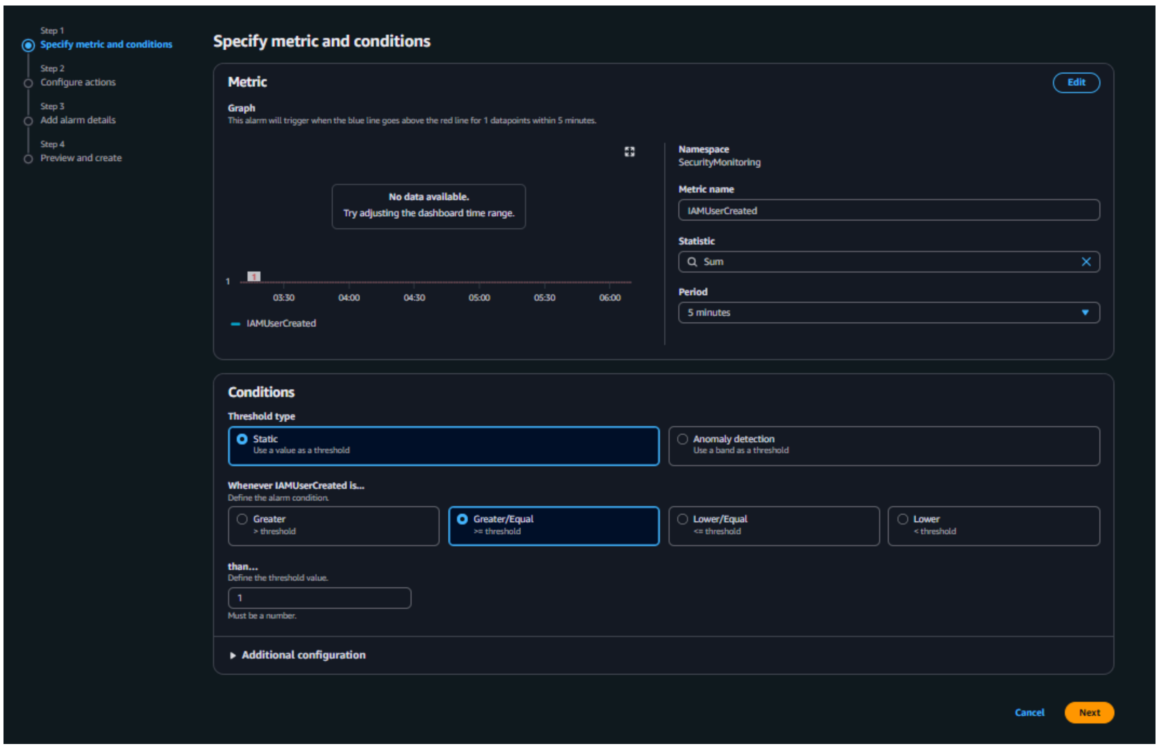


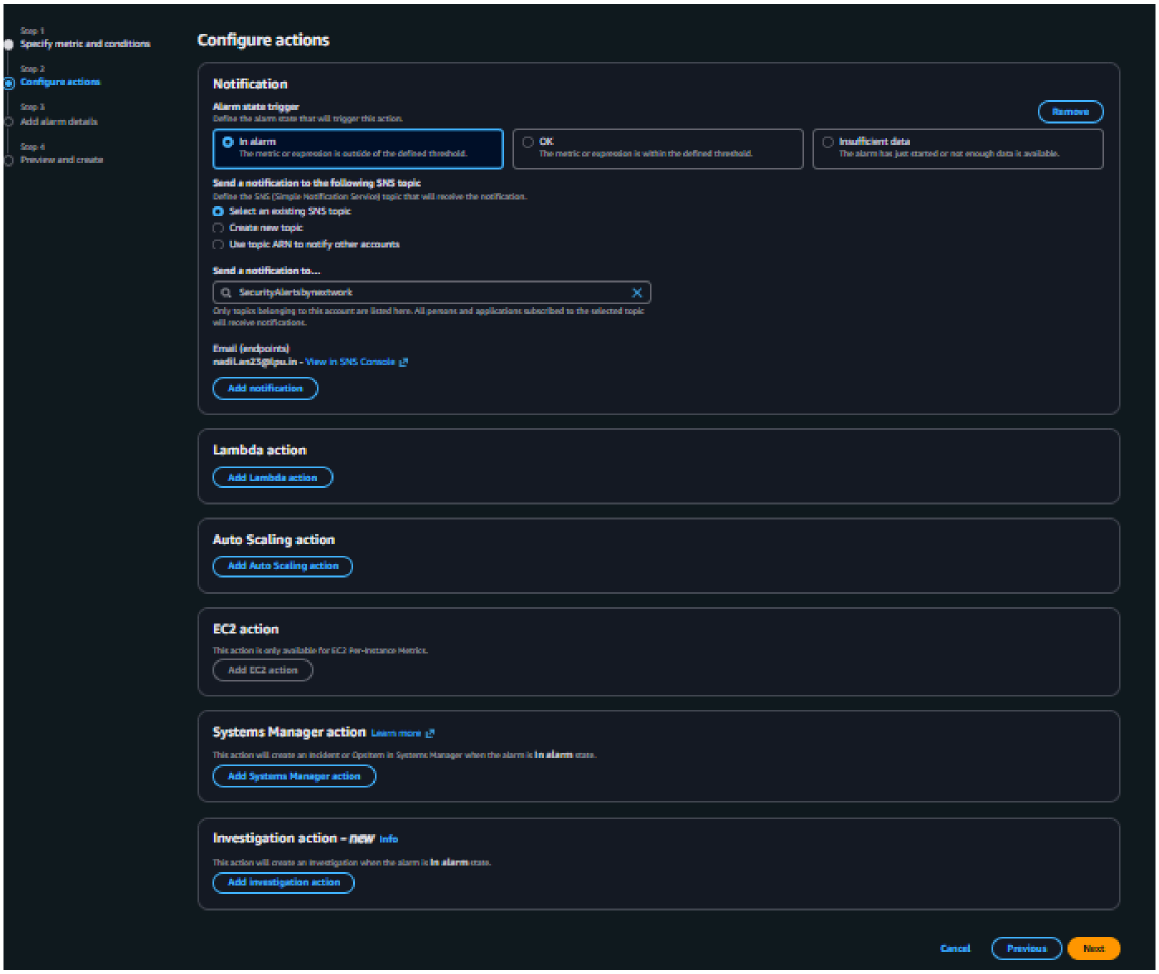


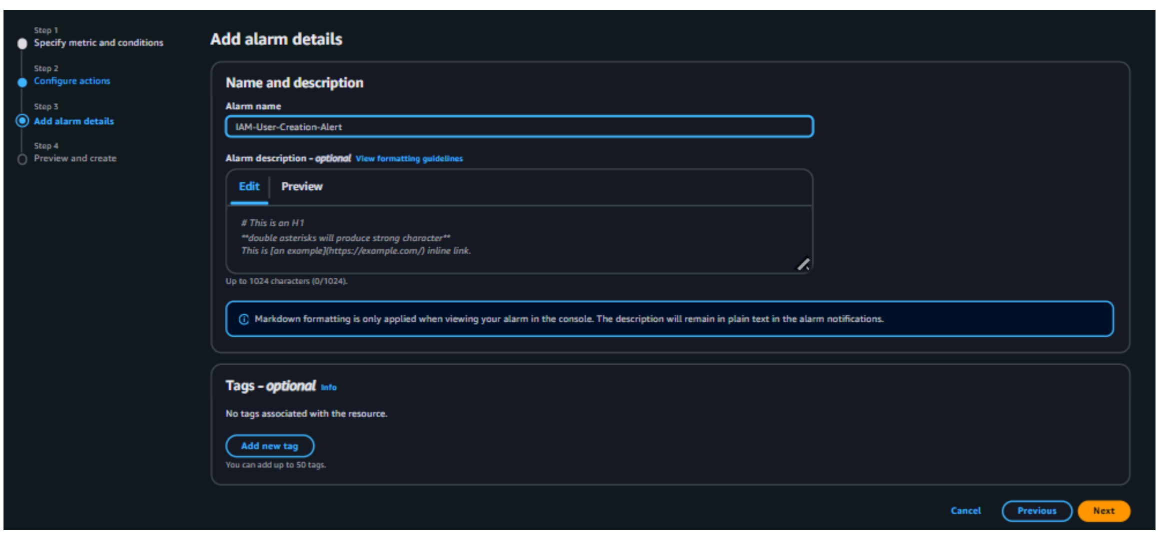
Step 6: Create CloudWatch Alarm

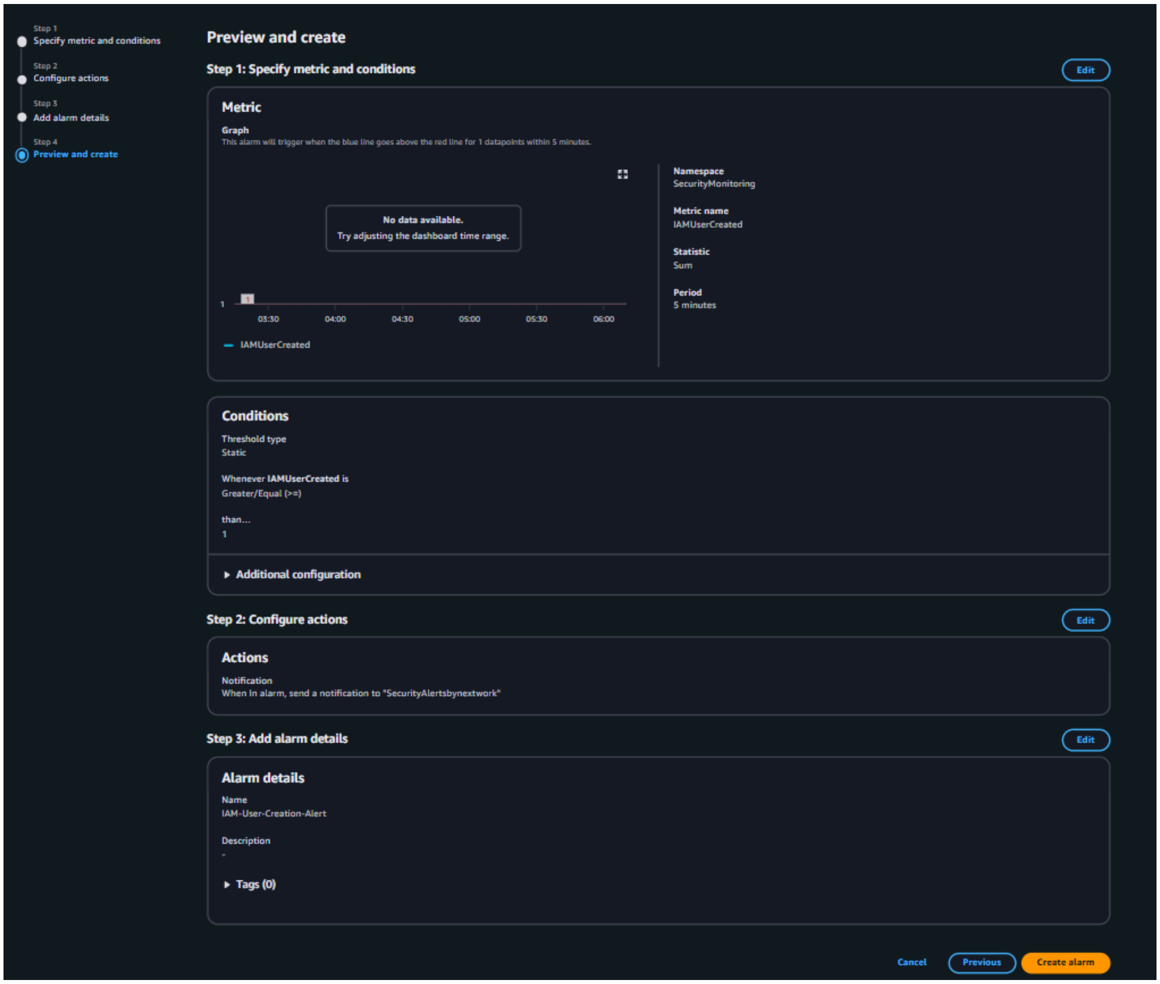
Create a CloudWatch Alarm with Statistic=Sum, Threshold >=1, and Period=5 minutes to detect CreateUser events.

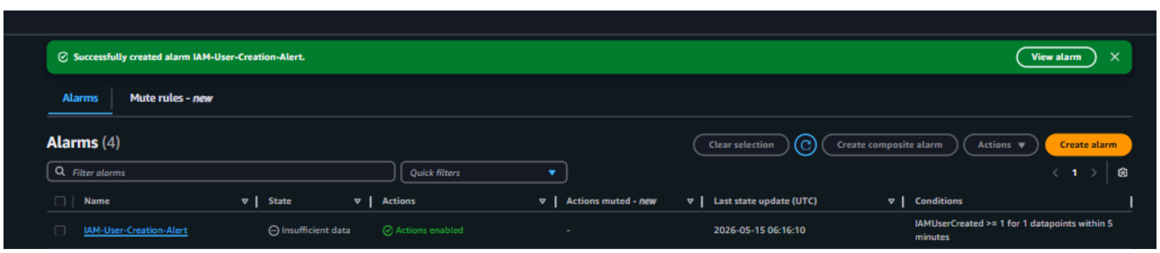





Step 1
Specify metric and conditions
Step 2
Configure actions
Step 3
Add alarm details
Step 4
Preview and create
Configure actions
Notification
Alarm state trigger
Define the alarm state that will trigger this action.
Remove
In alarm
The metric or expression is outside of the defined threshold.
OK
The metric or expression is within the defined threshold.
Insufficient data
The alarm has just started or not enough data is available.
Send a notification to the following SNS topic
Define the SNS (Simple Notification Service) topic that will receive the notification.
Select an existing SNS topic
Create new topic
Use topic ARN to notify other accounts
Send a notification to...
SecurityAlertsbynextwork
Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.
Email (endpoints)
View in SNS Console
Add notification
Lambda action
Add Lambda action
Auto Scaling action
Add Auto Scaling action
EC2 action
This action is only available for EC2 Per-Instance Metrics.
Add EC2 action
Systems Manager action
Learn more
This action will create an incident or OpsItem in Systems Manager when the alarm is In alarm state.
Add Systems Manager action
Investigation action - new
Info
This action will create an investigation when the alarm is In alarm state.
Add investigation action
Cancel
Previous
Next

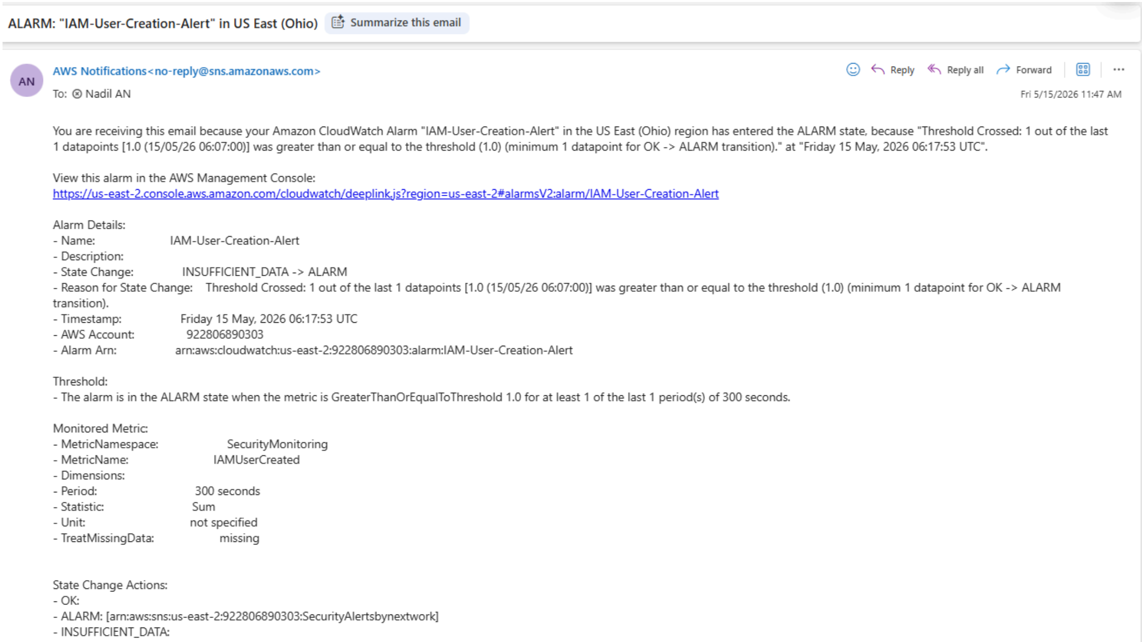

Step 1
Specify metric and conditions
Step 2
Configure actions
Step 3
Add alarm details
Step 4
Preview and create
Add alarm details
Name and description
Alarm name
IAM-User-Creation-Alert
Alarm description - optional
View formatting guidelines
Edit
Preview
This is an H1
double asterisks will produce strong character
This is [an example](https://example.com/) inline link.
Up to 1024 characters (0/1024).
Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.
Tags - optional
Info
No tags associated with the resource.
Add new tag
You can add up to 50 tags.
Cancel
Previous
Next





Step 9: Verify SNS Alert & Splunk Detection

Confirm SNS email alert delivery and validate CreateUser event ingestion into Splunk using SPL query: index=* sourcetype=aws:cloudtrail eventName=CreateUser



ALARM: "IAM-User-Creation-Alert" in US East (Ohio) Summarize this email

AN AWS Notifications<no-reply@sns.amazonaws.com> Reply Reply all Forward

To: Nadil AN Fri 5/15/2026 11:47 AM

You are receiving this email because your Amazon CloudWatch Alarm "IAM-User-Creation-Alert" in the US East (Ohio) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [1.0 (15/05/26 06:07:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Friday 15 May, 2026 06:17:53 UTC".

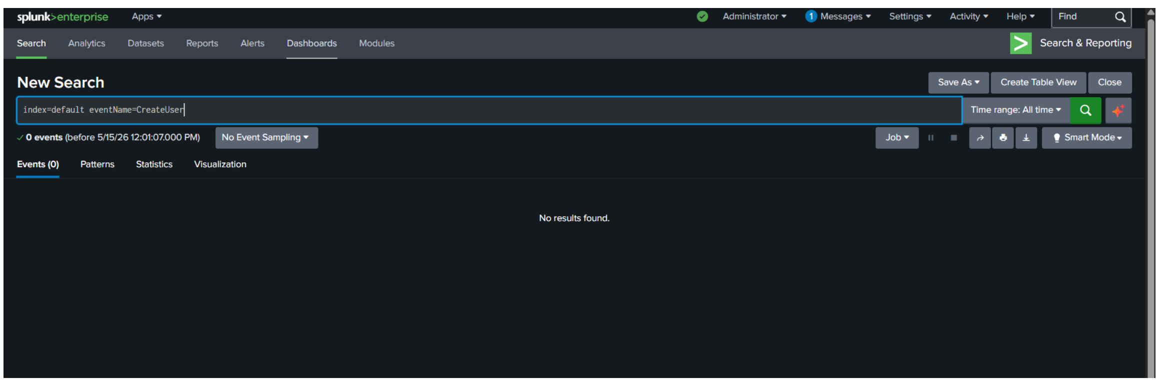
View this alarm in the AWS Management Console:
https://us-east-2.console.aws.amazon.com/cloudwatch/deeplink.js?region=us-east-2#alarmsV2:alarm/IAM-User-Creation-Alert

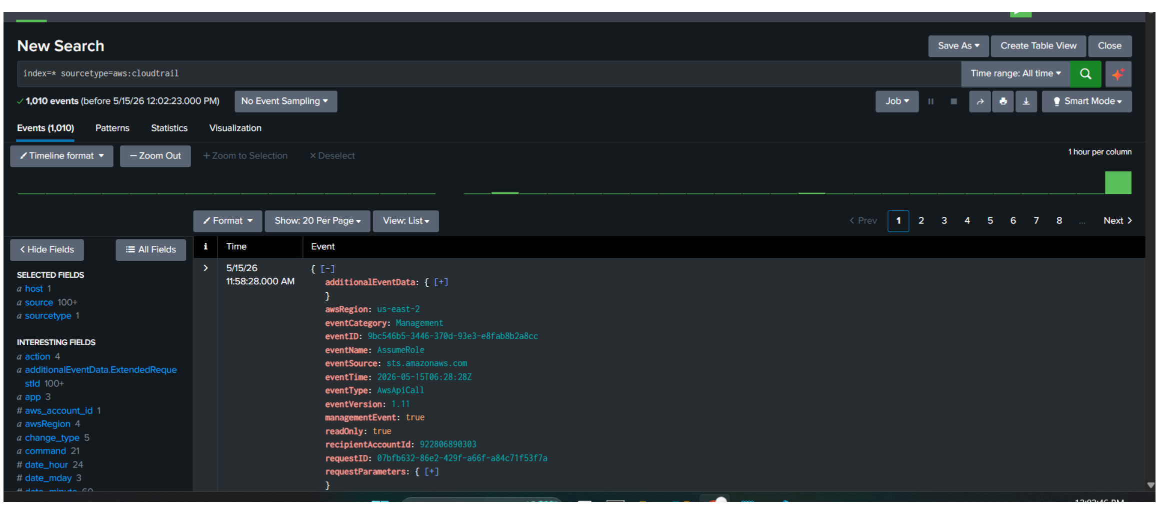
Alarm Details:
- Name: IAM-User-Creation-Alert
- Description:
- State Change: INSUFFICIENT_DATA -> ALARM
- Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [1.0 (15/05/26 06:07:00)] was greater than or equal to the threshold (1.0) (minimum 1 datapoint for OK -> ALARM transition).
- Timestamp: Friday 15 May, 2026 06:17:53 UTC
- AWS Account: 922806890303
- Alarm Arn: arn:aws:cloudwatch:us-east-2:922806890303:alarm:IAM-User-Creation-Alert

Threshold:
- The alarm is in the ALARM state when the metric is GreaterThanOrEqualToThreshold 1.0 for at least 1 of the last 1 period(s) of 300 seconds.

Monitored Metric:
- MetricNamespace: SecurityMonitoring
- MetricName: IAMUserCreated
- Dimensions:
- Period: 300 seconds
- Statistic: Sum
- Unit: not specified
- TreatMissingData: missing

State Change Actions:
- OK:
- ALARM: [arn:aws:sns:us-east-2:922806890303:SecurityAlertsbynextwork]
- INSUFFICIENT_DATA:





Conclusion

Successfully implemented AWS SOC monitoring for IAM user creation activities using CloudTrail, CloudWatch Metric Filters, CloudWatch Alarms, SNS notifications, S3 log storage, SQS integration, and Splunk SIEM. This workflow demonstrates real-time detection engineering, alerting, and centralized security monitoring for suspicious AWS IAM activities.